

FermiForge radial iteration diagnostics

Methods, measured performance and residual structure

2 October 2026

1 Problem, conventions and diagnostic definitions

This note documents BB, fixed-step Polyak momentum and restarted Anderson iteration for the same symmetry-constrained self-consistency problem. Methods are specified first, followed by measured results and comparison. The purpose is to distinguish accelerator behaviour from physical vortex structure and from discretization error.

The five comparisons start from identical archived initial field arrays: a compact C_{0+} seed on a phase-wound B-phase background, at $T/T_c = 0.30$ and $F_1^s = 5.4$. There are 60 independent radial nodes, reconstructed on a 119×119 mesh, with axial symmetry and translation invariance along the vortex axis. The radial domain extends to $60\xi_0$; the evolving asymptotic fit uses $45\xi_0 < r < 55\xi_0$, with continuation to $120\xi_0$. All runs use quadratic interpolation, trajectory step limit $0.25\xi_0$, eight generated Ozaki poles at cutoff 50, and ten MPI ranks. The generated bulk gap is 0.2798948241 in solver units.

Let $x_k \in \mathbb{R}^{1260}$ contain real and imaginary parts of the nine order-parameter components and three real Fermi-liquid fields at each node. For the discrete self-consistency map F , define

$$r_k = F(x_k) - x_k, \quad u_k = x_{k+1} - x_k, \quad (1)$$

$$E_{\text{rms}}(k) = \sqrt{r_k^T r_k / 1260}, \quad E_{\text{max}}(k) = \max_j |r_{k,j}|. \quad (2)$$

Convergence uses E_{max} , not update magnitude. Regional norms restrict these definitions to $r \leq 3\xi_0$, $3\xi_0 < r \leq 15\xi_0$, and $r > 15\xi_0$ (10, 23 and 27 nodes). These are diagnostic bins, not measured hard/soft-core boundaries. Norms are unweighted by area; squared-norm fractions are neither physical energies nor area fractions.

Signed files retain x_k, r_k, u_k . Reconstructed norms and state continuity $x_{k+1} = x_k + u_k$ are checked for BB and all three Anderson cases, including the restart boundary. At accepted convergence no update is applied. Polyak was run without signed recording, so only scalar/region histories are analysed. A small fixed-point residual does not bound interpolation or quadrature error; refinement comparisons are required for those errors.

2 Iteration methods and parameters

2.1 Barzilai–Borwein (BB)

With $s_k = x_k - x_{k-1}$ and $y_k = r_{k-1} - r_k$, the signed proposals are

$$\alpha_k^{\text{BB1}} = \frac{s_k^T s_k}{s_k^T y_k}, \quad \alpha_k^{\text{BB2}} = \frac{s_k^T y_k}{y_k^T y_k}, \quad x_{k+1} = x_k + \alpha_k r_k. \quad (3)$$

The run alternates BB1/BB2, starts at $\alpha = 1$, clips to $[0.001, 100]$, uses absolute curvature and disables growth damping. Curvature affects the step rule; residual alignment tests whether successive residual vectors point in similar directions. They are different diagnostics.

2.2 Polyak momentum

The implementation follows SuperConga Appendix E, E4/E6/E7 [1]:

$$v_{k+1} = (1 - \beta)v_k + \alpha r_k, \quad x_{k+1} = x_k + v_{k+1}, \quad v_0 = 0. \quad (4)$$

The tested fixed parameters are $\alpha = 2$, $\beta = 0.5$. This is heavy-ball momentum, not objective-value-based Polyak step sizing. The accumulated update is not generally parallel to the current residual.

2.3 Restarted Anderson (AA)

The implementation translates the legacy residual-minimizing iterator. From retained (x_i, r_i) , choose the smallest-residual pair (x_b, r_b) and form columns $D_i = r_i - r_b$, $S_i = x_i - x_b$. Then

$$c = \arg \min_c \|r_b + Dc\|_2^2, \quad D^T Dc = -D^T r_b, \quad (5)$$

$$x_{k+1} = x_b + Sc + p_k(r_b + Dc). \quad (6)$$

The code solves normal equations and selects/discards history by residual norm and a progress threshold. The adaptive rule for p_k is specified below; p_k multiplies the minimized residual, not generally r_k . History capacity is 10 and the progress threshold is 0.1. Every 20 maps history is cleared and p resets to 0.01; no simple-update cycles intervene. Completed caps are 3, 5 and 10.

2.4 How the Anderson mixing parameter is chosen

The following is the implemented legacy rule in `src/legacy_anderson_mixing.f90`, not a universal Anderson prescription. Let p_k^{old} be the stored mixing value on entry, and M_k the stored *squared norm* of the preceding linear residual prediction. With $r_k = F(x_k) - x_k$ the newly evaluated residual, compute

$$\eta_k = \min\left(1, \sqrt{\max\left(\frac{M_k}{\|r_k\|_2^2}, 0\right)}\right). \quad (7)$$

This factor is computed before discarding/reordering history. Next choose the smallest-residual reference among the retained points, solve for c , and form the displacement $d_k = Sc$ and predicted residual $\tilde{r}_k = r_b + Dc$. For a successful solve with nonempty history,

$$p_k^{\text{trial}} = \frac{\eta_k}{2} \left(p_k^{\text{old}} + \frac{\|d_k\|_2}{\|r_b\|_2} \right), \quad (8)$$

$$p_k = \max\left(0.01, \min(p_{\text{max}}, p_k^{\text{trial}})\right), \quad (9)$$

$$x_{k+1} = x_b + d_k + p_k \tilde{r}_k, \quad M_{k+1} = \|\tilde{r}_k\|_2^2. \quad (10)$$

All norms are the same unweighted Euclidean vector norms used by the iterator. Crucially, η_k uses the *current measured* residual r_k , whereas the displacement ratio uses the *smallest retained* residual r_b , which may belong to an older iterate. Also, $\tilde{r}_k$ is a linear combination of saved residuals, not a fresh evaluation of $F(x_b + d_k)$ or $F(x_{k+1})$.

Interpretation. The displacement-to-residual ratio supplies a candidate mixing scale, averaged with the previous value. If the newly measured residual exceeds the preceding prediction in norm, $\eta_k < 1$ reduces the proposal. Otherwise $\eta_k = 1$. For example, a displacement ratio of 8 and stored $p = 2$ give a trial value 5 when $\eta = 1$, or 2.5 when $\eta = 1/2$. The cap and floor are then applied. This is a heuristic adaptation, not minimization over p , a free-energy line search or a guarantee that the next residual decreases. The threshold 0.1 controls history selection; it is not an extra multiplier in this formula.

Special cases. A reset empties history, sets $p = 0.01$ and $M = 0$. The first unconverged call then uses $x_{k+1} = x_k + 0.01r_k$ and stores $M_{k+1} = \|r_k\|_2^2$; it bypasses the adaptive formula. If convergence is already accepted, the state is left unchanged and the reported p is the stored value, not an applied step. If the coefficient solve fails, the code instead takes its sinusoidal fallback step, stores $p = 1$ and $M = 10^{29}$, and flags the fallback; this did not occur in the analysed Anderson runs.

2.5 Stopping and restart protocol

BB and Polyak target $E_{\text{max}} \leq 2 \times 10^{-6}$. AA5 first stops at 2×10^{-5} after 80 maps, then restarts from its saved fields at the tighter tolerance. Fresh history coincides with a scheduled AA20 boundary. The first continuation map repeats the accepted state's evaluation; this is not an identical uninterrupted-trajectory claim. AA10 uses the tight tolerance from scratch, as does AA3. Anderson segments permit at most 600 maps and record checkpoints and signed diagnostics each map.

3 Measured performance of each method

3.1 BB

BB converges at map 433 with RMS/max 4.0912×10^{-7} / 1.9427×10^{-6} . Its maximum-residual history exactly reproduces the earlier scratch BB run. All 432 observed secant products $s_k^T y_k$ are positive, so absolute curvature does not alter their sign here. There are 37 capped updates, 159 RMS increases and 124 negative consecutive-residual cosines. Of 65 updates followed by an RMS doubling, 50 use BB1 and 15 BB2. Update 329 has $\alpha = 21.05$, below the cap, yet is followed by a 17.25-fold RMS increase. The cap is not the sole source of spikes.

Final inner/transition/outer squared-residual fractions are 0.82%, 23.99% and 75.18%. The largest component is $\text{Re } A_{xz}$ at $44.13\xi_0$, just inside the fit annulus; location alone does not establish a boundary defect. Complete numerical output is available, but the manifest records the previously observed MPI shutdown failure.

3.2 Polyak

Polyak reaches its 600-map limit with RMS/max 1.9761×10^{-6} / 8.4387×10^{-6} , not the requested maximum-error tolerance. RMS increases only once. Final regional RMS values, from inner to outer, are 4.90×10^{-7} , 1.81×10^{-6} and 2.41×10^{-6} . The launcher exits cleanly. Signed spatial-pattern claims cannot be made because the needed vectors were not saved.

3.3 Anderson, $p_{\max} = 3$

This scratch run converges cleanly at map 154, with RMS/max $3.9395 \times 10^{-7} / 1.7961 \times 10^{-6}$. Its 153 applied updates include 122 at the cap. There are ten RMS increases, one RMS doubling, no negative full-residual cosines and no stochastic fallbacks. No residual-direction reversal does not imply monotonic error: magnitude and direction are distinct. Final regional RMS values are 3.0394×10^{-7} , 3.8794×10^{-7} and 4.2717×10^{-7} ; squared-norm fractions are 9.92%, 37.17% and 52.91%. The largest component is $\text{Re } A_{xz}$ at $54.17\xi_0$, in the fitting annulus. Location alone does not establish a boundary defect. Signed norms, state continuity and equality of the initial field arrays with the other scratch runs have been checked.

3.4 Anderson, $p_{\max} = 5$

The continuation needs 81 additional maps: 161 recorded evaluations including the repeated restart evaluation, and 159 applied updates. Final RMS/max are $2.9688 \times 10^{-7} / 1.9918 \times 10^{-6}$. There are 38 RMS increases, no doublings, four negative full-residual cosines, 105 updates at the cap and no stochastic fallbacks. Both segments exit cleanly. Final regional RMS values are 3.1981×10^{-7} , 2.5410×10^{-7} and 3.2090×10^{-7} ; squared-norm fractions are 19.34%, 28.08% and 52.58%. The maximum is in the y Fermi-liquid field at $0.305\xi_0$.

3.5 Anderson, $p_{\max} = 10$

This scratch run converges cleanly at map 224: RMS/max $4.6496 \times 10^{-7} / 1.9652 \times 10^{-6}$. Its 223 applied updates include 112 at the cap. There are 90 RMS increases, three doublings, 18 negative full-residual cosines and no stochastic fallbacks. Final regional RMS values are 1.6769×10^{-7} , 5.1106×10^{-7} and 4.9751×10^{-7} ; squared-norm fractions are 2.17%, 46.31% and 51.52%. The maximum is $\text{Re } A_{xz}$ at $35.93\xi_0$. Different limiting components do not by themselves imply different physical solutions.

4 Spatial residual structure: definition and illustration

4.1 Physical texture versus numerical residual

The hard-core/soft-core distinction concerns the solution itself: strong changes of the order parameter near the hard core, and a spatial rotation texture of the 3×3 matrix in the outer core. A useful local description in a B-like region is

$$A(\mathbf{r}) \simeq \Delta(\mathbf{r})e^{i\phi(\mathbf{r})}R_{\text{so}}(\mathbf{r}) + \text{non-bulk corrections.} \quad (11)$$

This texture can persist after convergence. The residual $F(A) - A$ instead measures the change requested by the discrete self-consistency map. A rotation-like residual suggests incomplete adjustment of orientation; it is not the rotation texture itself, nor a physical-time oscillation.

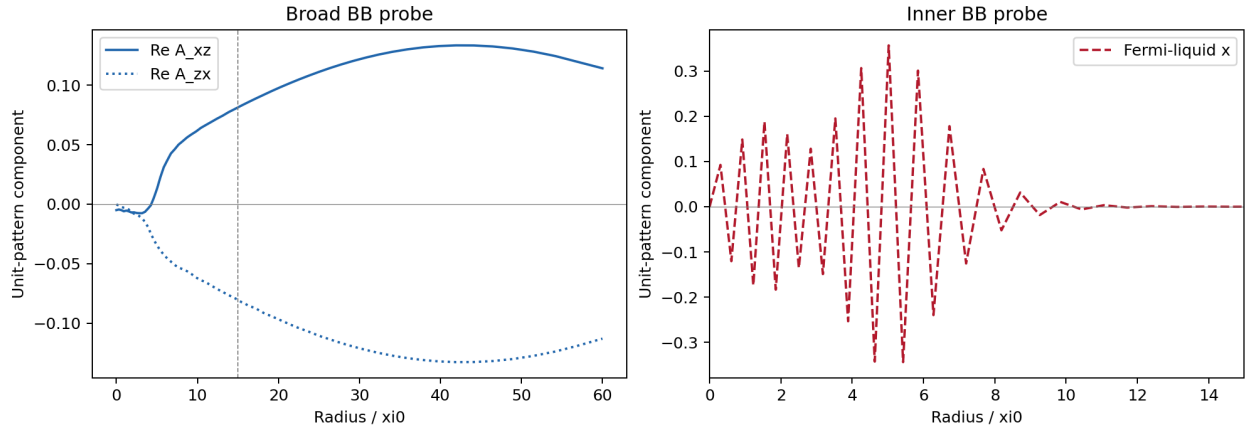
For the spin-first convention, test the geometric tangent to a spin rotation about y of the final field:

$$T_y = L_y A, \quad L_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad \delta A = \delta\theta_y T_y. \quad (12)$$

Pack real/imaginary parts into the same layout and set Fermi-liquid entries to zero. The overlap on $r > 15\xi_0$ is a shape test. It neither proves a free rotation symmetry nor measures a dipole length, Hessian eigenvalue or permissible zero mode. No such direction has been projected out.

4.2 Two fixed spatial probes

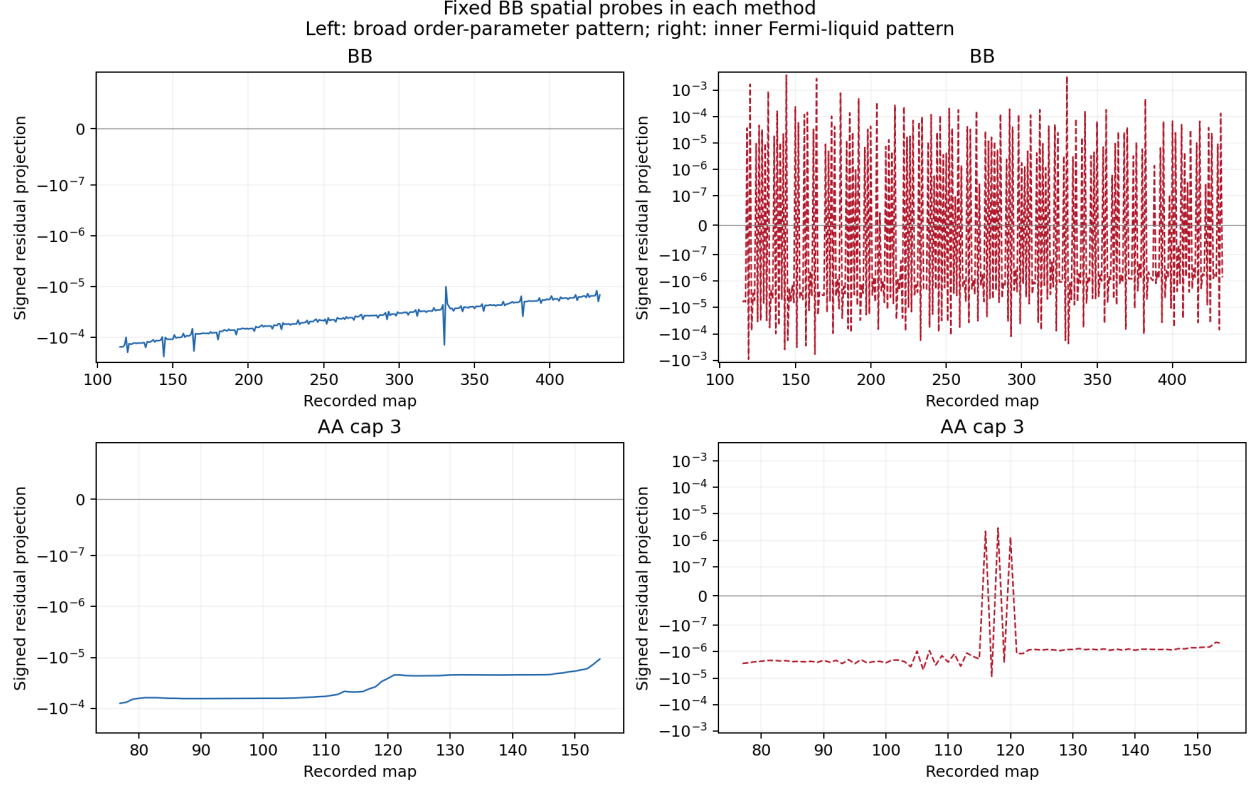
The earlier BB analysis used an uncentred SVD of unit residual vectors $\hat{r}_k = r_k/\|r_k\|$ at maps 251–433. Its first two unit patterns b_1, b_2 account for 61.77% and 19.49% of that matrix's squared norm. The first is broad in $\text{Re } A_{xz}, \text{Re } A_{zx}$, with outer rotation-tangent cosine 0.9923. The second has 99.2% of its norm squared in the x Fermi-liquid component. These are empirical patterns, not Jacobian eigenvectors. Their signs are arbitrary but held fixed below.



Left: opposing off-diagonal components of the broad pattern. Right: magnified $r \leq 15\xi_0$, showing spatial sign alternation as well as localization. This fine spatial structure should not be identified with a smooth physical core mode without a refinement test.

4.3 Evolution under BB and Anderson

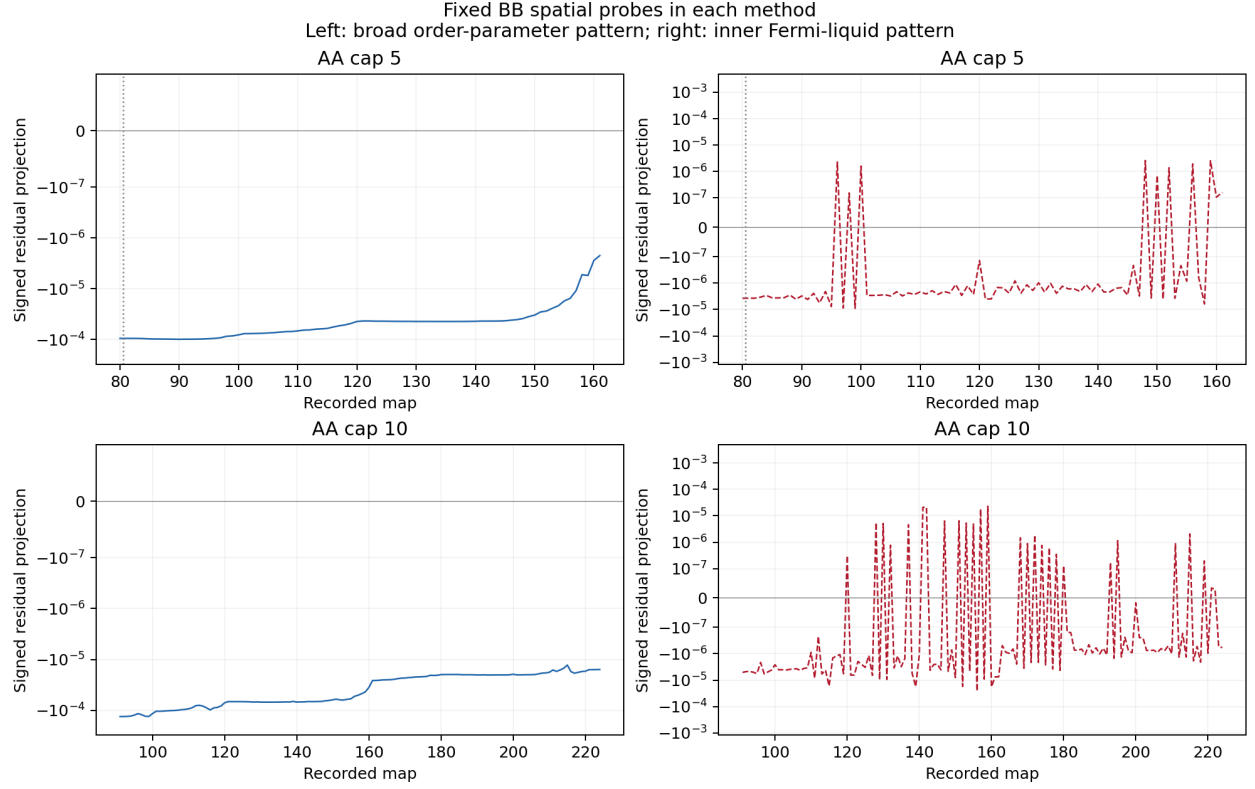
Project each method’s *unnormalized* residual onto the same two BB probes: $q_j(k) = b_j^T r_k$. These plots retain absolute decay, unlike projections of unit residuals. They start at the first $E_{\max} \leq 2 \times 10^{-5}$ crossing and include all subsequent maps: BB 115–433, AA3 77–154, AA5 80–161, AA10 91–224. Windows have different lengths and do not represent identical states. AA5 includes its repeated restart evaluation. BB-derived probes are a common ruler, not a neutral basis optimised for each method.



Reading the panels. Left: the broad projection keeps its sign and approaches zero, non-monotonically and at different rates. Right: the inner projection repeatedly crosses zero, especially for BB. This is what “oscillatory” means here: iteration-to-iteration residual sign reversals. Vertical scales match within each column across both figure pages and are symmetric-logarithmic (linear for $|q| < 10^{-7}$); horizontal ranges differ. Inner projections are much smaller for Anderson, but their sign changes have not disappeared.

4.4 Signed histories for the larger Anderson caps

The same fixed BB probes and vertical scales are used here. The dotted vertical line marks AA5's restart; its first continuation evaluation repeats the previously accepted state. Neither pattern is removed by Anderson.



The AA3 and AA5 inner projections are small relative to BB's, whereas AA10 has more visible excursions. These projections sample one fixed inner shape; they do not include every possible spatial pattern of the inner residual.

4.5 Independent checks of the Anderson tendencies

To avoid relying only on a BB-derived basis, repeat the uncentred unit-residual SVD independently for each method over the threshold-defined windows.

Diagnostic in threshold window	BB	AA3	AA5	AA10
Leading-pattern squared-norm share	61.51%	80.36%	79.97%	70.40%
Leading pattern: outer rotation cosine	0.9957	0.9904	0.9969	0.9934
Mean share along fixed broad BB probe	61.17%	78.41%	78.50%	70.13%
Mean share along fixed inner BB probe	17.47%	0.15%	0.27%	0.62%
Inner BB-probe sign changes	234	6	15	50
Inner FL-vector reversals / pairs	227/318	54/77	66/81	104/133

The mean fixed-probe share is $\langle (b_j^T r_k)^2 / \|r_k\|^2 \rangle_k$, not a physical energy. “Inner FL” includes all three Fermi-liquid components at $r \leq 15\xi_0$; a reversal means their consecutive-vector cosine is negative. Pattern shares depend on window, scaling and normalization. The SVD percentages here differ from the earlier maps-251–433 BB values for that reason.

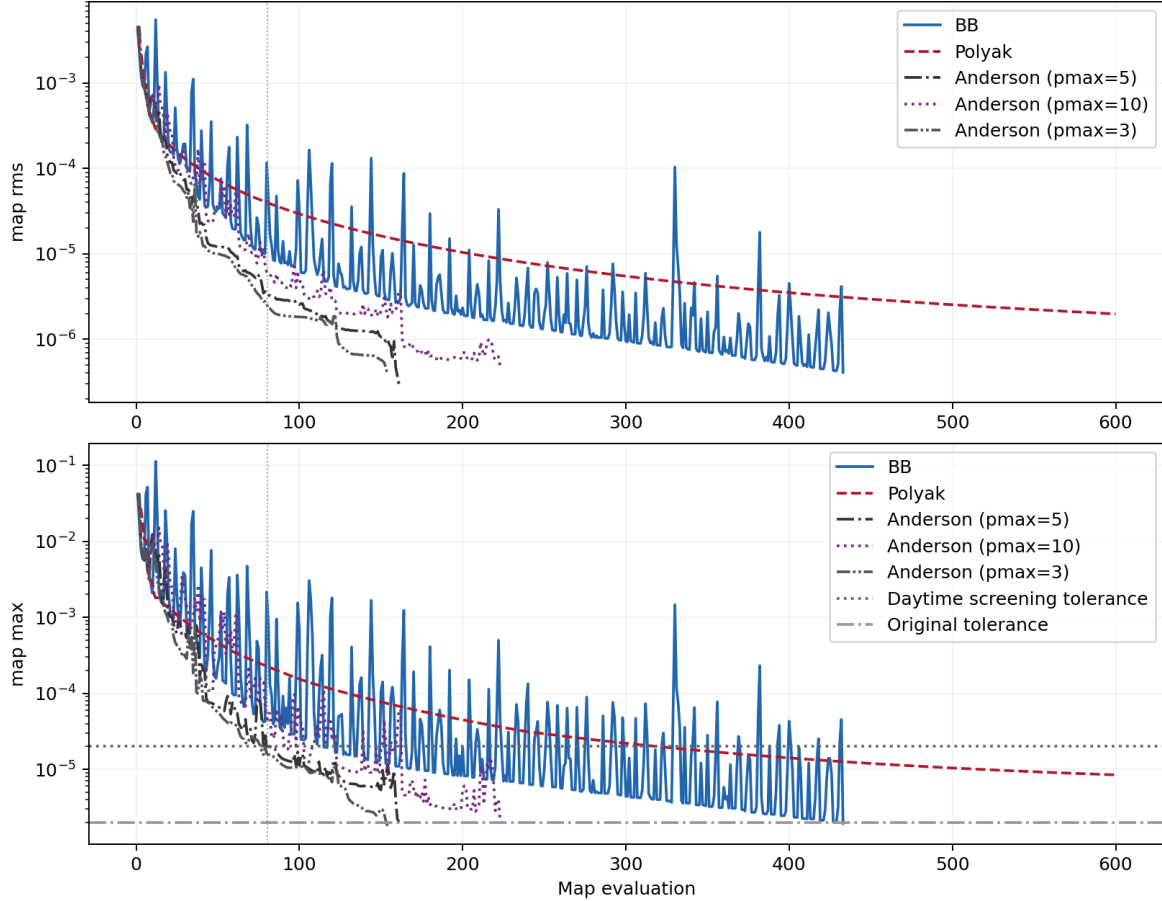
All four independent leading patterns resemble the outer rotation tangent. The rotation-like tendency is therefore not confined to BB. AA5’s second pattern is mixed (11.2% in inner FL); AA10’s second is mainly inner FL (92.4%), but carries only 4.37% of unit-residual content. BB’s second carries 17.89% and is 99.56% inner FL in this window. AA3’s second pattern carries 6.59% of unit-residual content and is predominantly outer (80.96%), with only 3.24% in inner FL.

The inner FL subvector actually reverses frequently in all Anderson runs. What is reduced is its weight in the *whole* residual, not necessarily its reversal frequency. A near-zero subvector can reverse without a large change in global error. This explains why four and eighteen full-vector reversals (and zero for AA3) do not imply an absence of inner oscillations. The fixed inner BB probe samples only one spatial shape; other inner patterns may remain.

These observations are consistent with orientation adjustment of the soft texture being a persistent numerical difficulty. They do not show that the accelerator creates the soft core, that the physical texture is unstable, or that iteration oscillations have a physical frequency. The earlier BB spectral peaks near 6.31, 8.71 and 10.17 maps describe iteration history, not an excitation spectrum or a single fixed clock.

5 Comparison and conclusions for this benchmark

First maximum-error crossing	BB	Polyak	AA3	AA5	AA10
2×10^{-4}	43	87	35	39	43
2×10^{-5}	115	319	77	80	91
10^{-5}	172	517	105	107	128
2×10^{-6}	433	not reached	154	161	224



AA3 has the fewest recorded maps of the tested settings: 154 versus 161 for AA5 and 224 for AA10. The improvement over staged AA5 is modest (4.3%); AA3 also crosses the loose threshold first, before AA5's restart affects the comparison. These samples do not establish an optimal cap. Polyak has the smoothest RMS history but has not reached the tight tolerance. This ranks these settings on this problem, not the algorithms universally.

Median map times are 17.34, 17.03, 17.08, 17.01 and 17.16 s for BB, Polyak, AA3, AA5 and AA10. Summed AA map times are 2635, 2761 and 3851 s. BB has substantial timing outliers, so total elapsed times are not controlled hardware measurements. AA5's staged execution remains a caveat; a tight single-segment control would remove it. AA3 is the best observed cap here, not a demonstrated universal optimum.

Structured residuals suggest testing history/reset policies and spatial resolution before invoking a numerical noise floor or removing a presumed zero mode. Physical hard/soft-core structure is the background to this interpretation, not something inferred solely from these residual plots.

6 Data provenance and reproduction

All directories below are relative to the FermiForge project root.

```
runs/261001-192802-radial-bb-diagnostics-0plus-T030-Fs54-radial-reference
runs/261001-233335-radial-polyak-scratch-0plus-T030-Fs54-radial-reference
runs/261002-091852-radial-anderson-aa20-diagnostics-0plus-T030-Fs54-radial-reference
runs/261002-094542-radial-anderson-aa20-0plus-pmax5-tight-continuation-radial-reference
runs/261002-111709-radial-anderson-aa20-0plus-pmax10-scratch-T030-Fs54-radial-reference
runs/261002-130520-radial-anderson-aa20-0plus-pmax3-scratch-T030-Fs54-radial-reference
```

Input namelists, metrics, histories, signed vectors and manifests are primary records. Signed layout is block-major: real and imaginary parts of A_{xx} , A_{xy} , $\dots$, A_{zz} , then three Fermi-liquid fields; each block has 60 nodes. Matrix indices are spin then orbital. No solver or archived run data were changed by this analysis.

To reproduce figures and numerical summaries:

```
python3 tools/analyze_signed_radial_bb.py \
    runs/261001-192802-radial-bb-diagnostics-0plus-T030-Fs54-radial-reference
python3 tools/compare_radial_accelerators.py
python3 tools/compare_signed_residual_patterns.py
cd docs/technical_note
pdflatex -interaction=nonstopmode accelerator_diagnostics.tex
```

Machine-readable summaries are retained in `work/signed-bb-analysis/`:

```
analysis.json
accelerator_comparison.json
signed_pattern_comparison.json
```

Figures are retained beside the source: compilation does not require raw data; regeneration needs the archived runs, NumPy and Matplotlib.

Scratch launchers create new dated outputs and start solver runs; they are not part of document reproduction:

```
bash tools/run_radial_anderson_diagnostics.sh 0plus --tolerance 2e-6
bash tools/run_radial_anderson_p10.sh
bash tools/run_radial_anderson_p3.sh
```

References

- [1] P. Holmvall et al., *SuperConga: An open-source framework for mesoscopic superconductivity*, Applied Physics Reviews **10**, 011317 (2023), doi:10.1063/5.0100324. Appendix E supplies the Polyak recurrence and defaults.